TREES AND FINITE DIFFERENCES

QTR Equities India

J.P.Morgan

Main Points

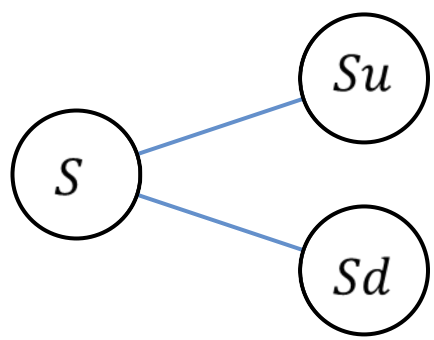
- *Binomial Trees*
- Trinomial Trees
- Explicit Finite Difference Schemes
- Implicit & Crank-Nicolson
- Barrier Options

Binomial Trees

- Basic SDE for stock in the BS model (no divs, no repo), in the risk-neutral measure

$$dS_t = rS_t dt + \sigma S_t dW_t$$

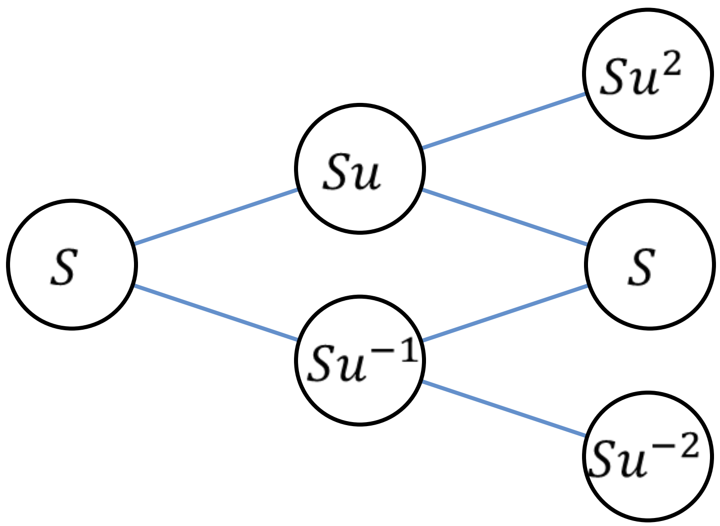
- Consider a discrete approximation of this process:
  - after a small time step the stock can go either up or down



TREES AND FINITE DIFFERENCES

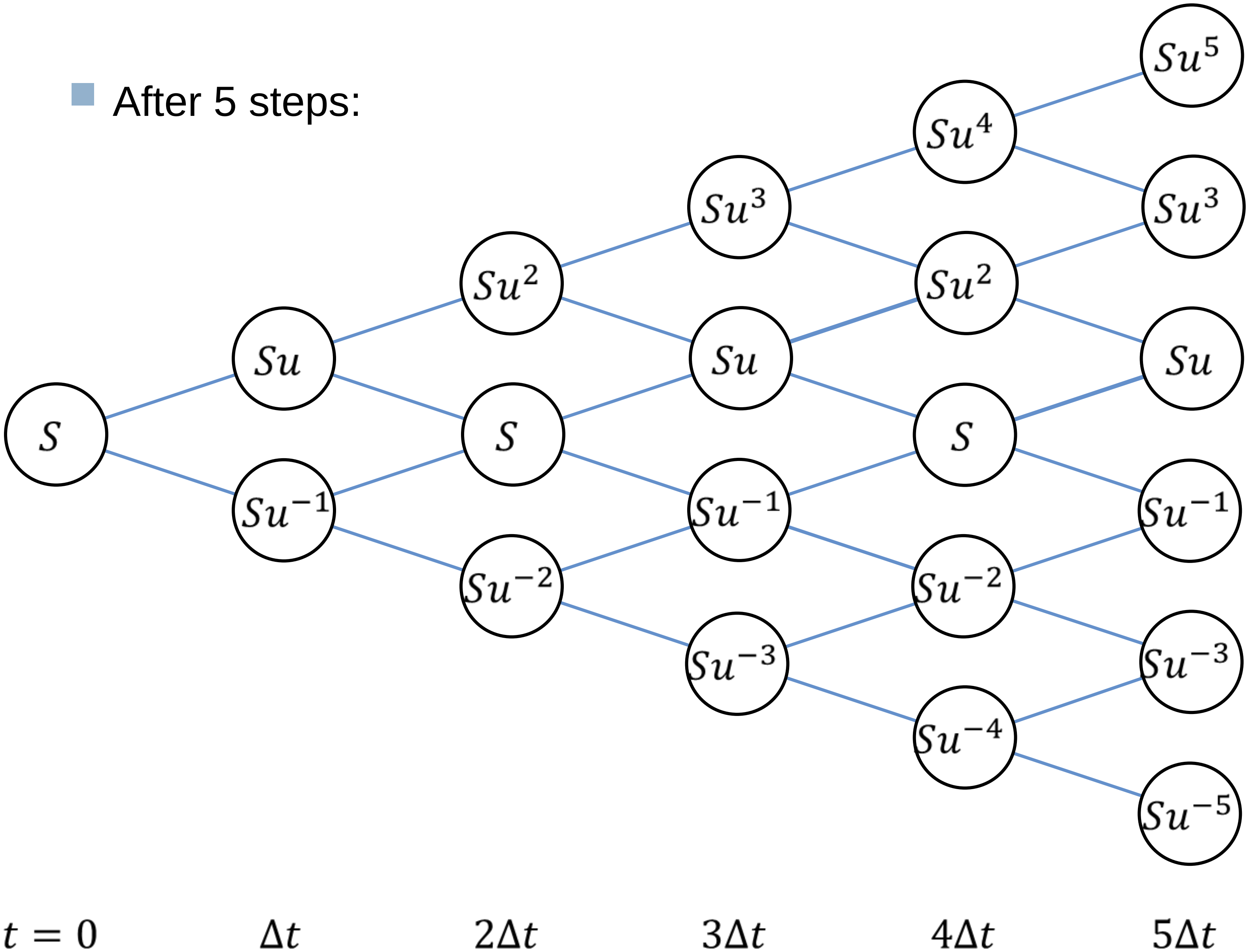
Binomial Trees

- If we take u and d such that $ud = 1$, then after two steps the tree recombines:



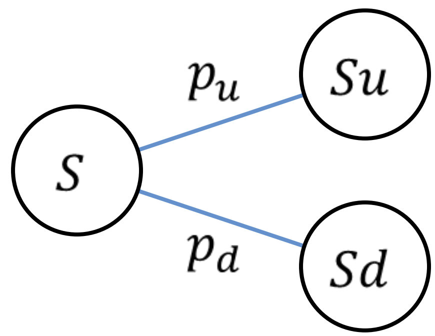
Binomial Trees

- After 5 steps:



Binomial Trees

- In the risk neutral measure all discounted tradable assets are martingales



- That means that $S = (p_u Su + p_d Sd)e^{-r\Delta t}$
- If we add the condition $p_u + p_d = 1$, we get a 2×2 system that can be solved

Binomial Trees

- The solution is

$$p_u = \frac{e^{r\Delta t} - d}{u - d}, \qquad p_d = \frac{u - e^{r\Delta t}}{u - d}$$

- The sizes of the up move and the down move are chosen to match the 2nd moment (i.e., the variance)

$$u = e^{\sigma \Delta t}$$

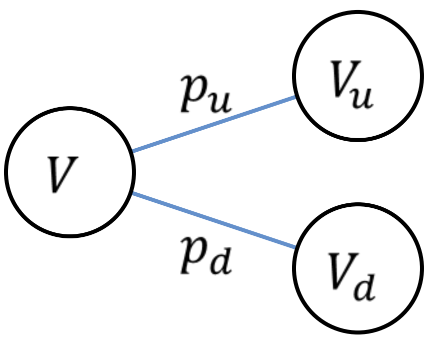
$$d = e^{-\sigma \Delta t}$$

- Check that

$$\mathrm{Var}[S(t + \Delta t) | S(t) = S] = \sigma^2 S^2 \Delta t + \text{higher order terms}$$

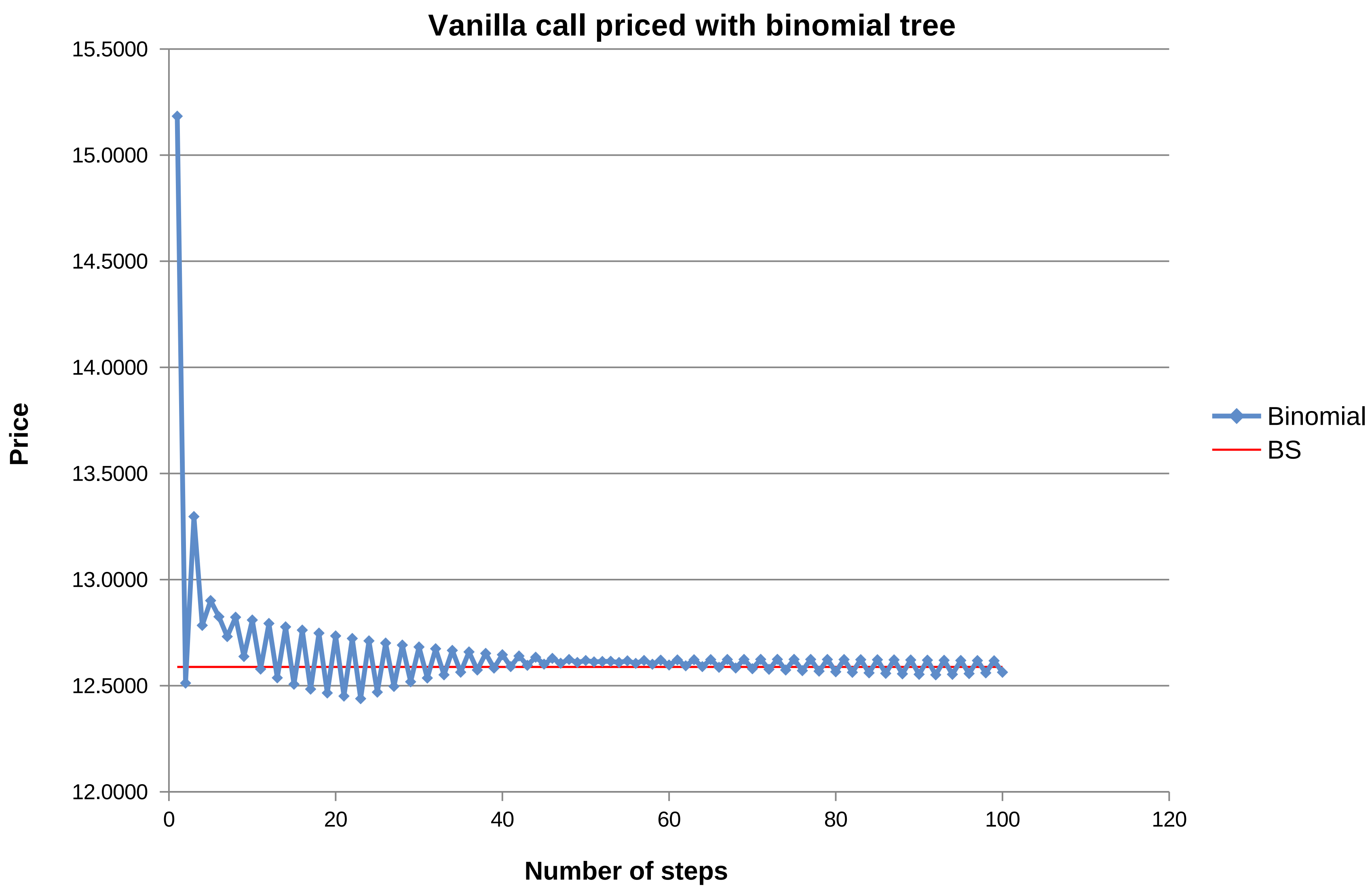
Binomial Trees

- Option pricing is done by backward induction
- Start at the last time step: the price of the option is known, it's simply the payoff
- Then proceed backward: at each node of the tree the price of the option is the calculated from the price of the option at the two nodes branching from it



$$V = e^{-r\Delta t}(p_u V_u + p_d V_d)$$

Binomial Trees



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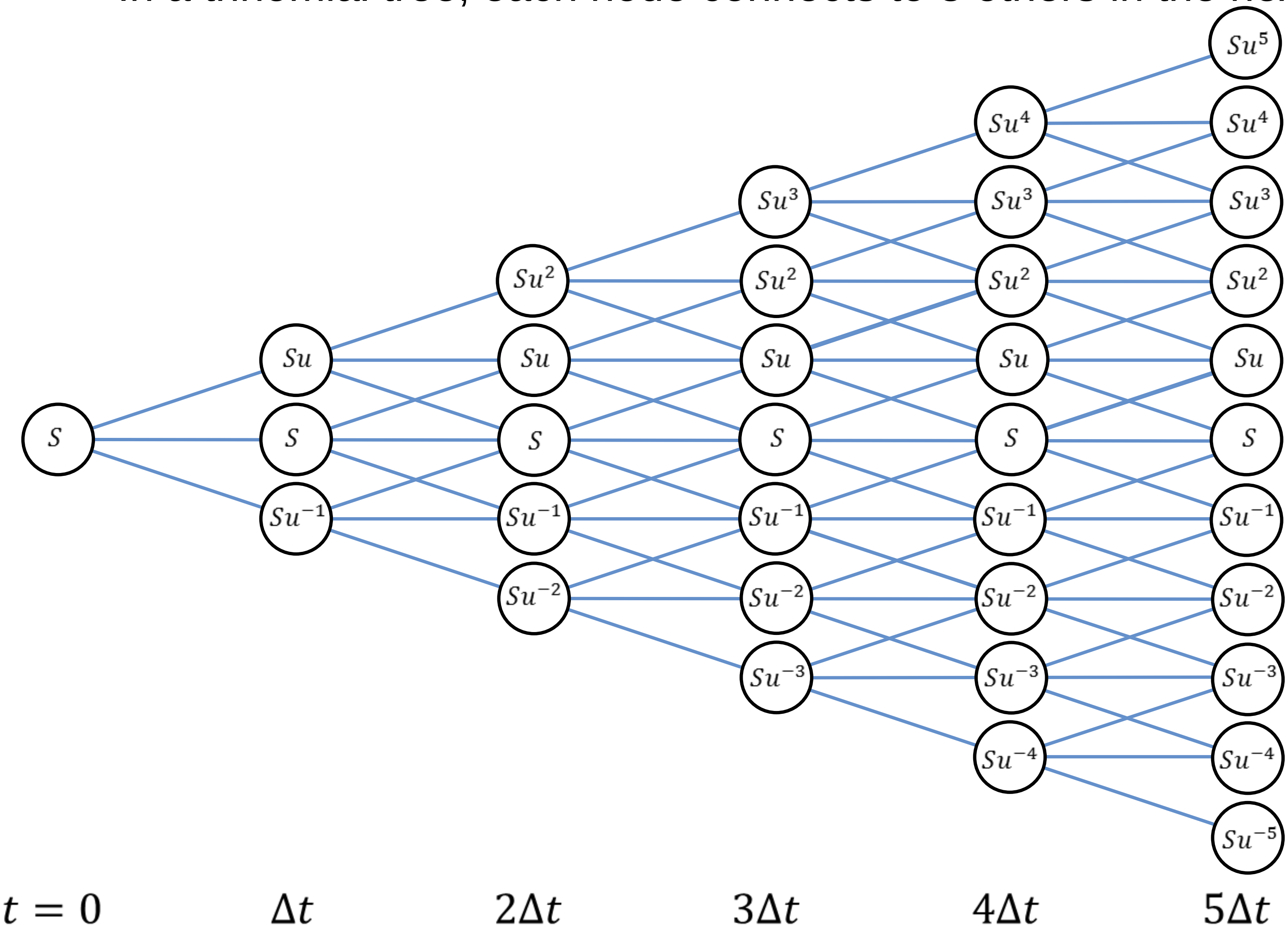
Trinomial Trees

- In a trinomial tree, each node connects to 3 others in the next step



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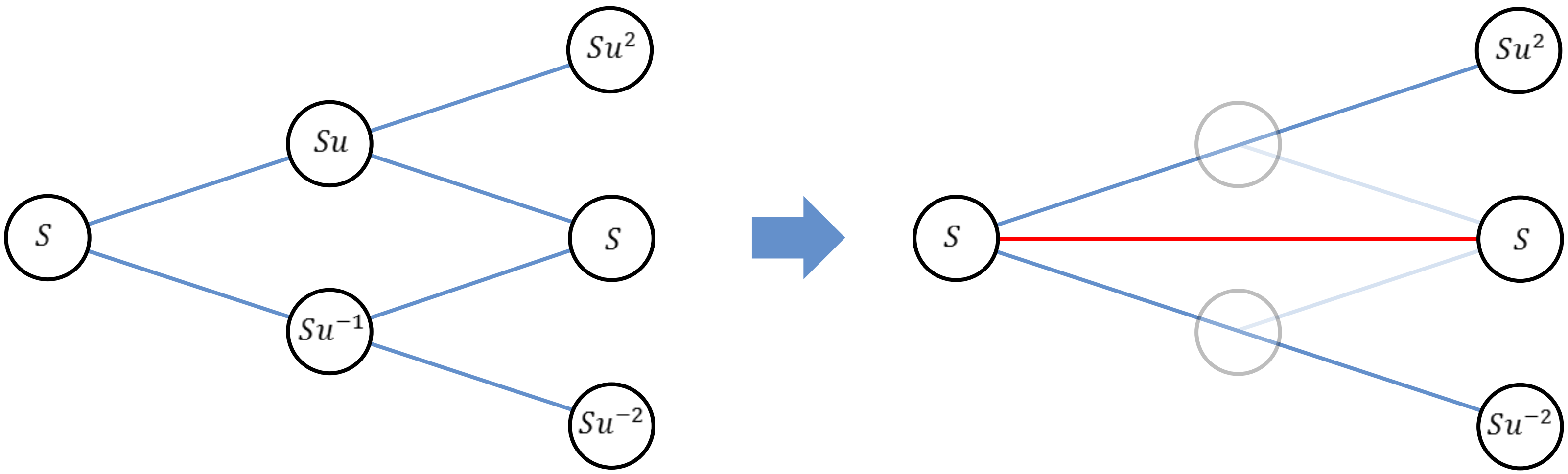
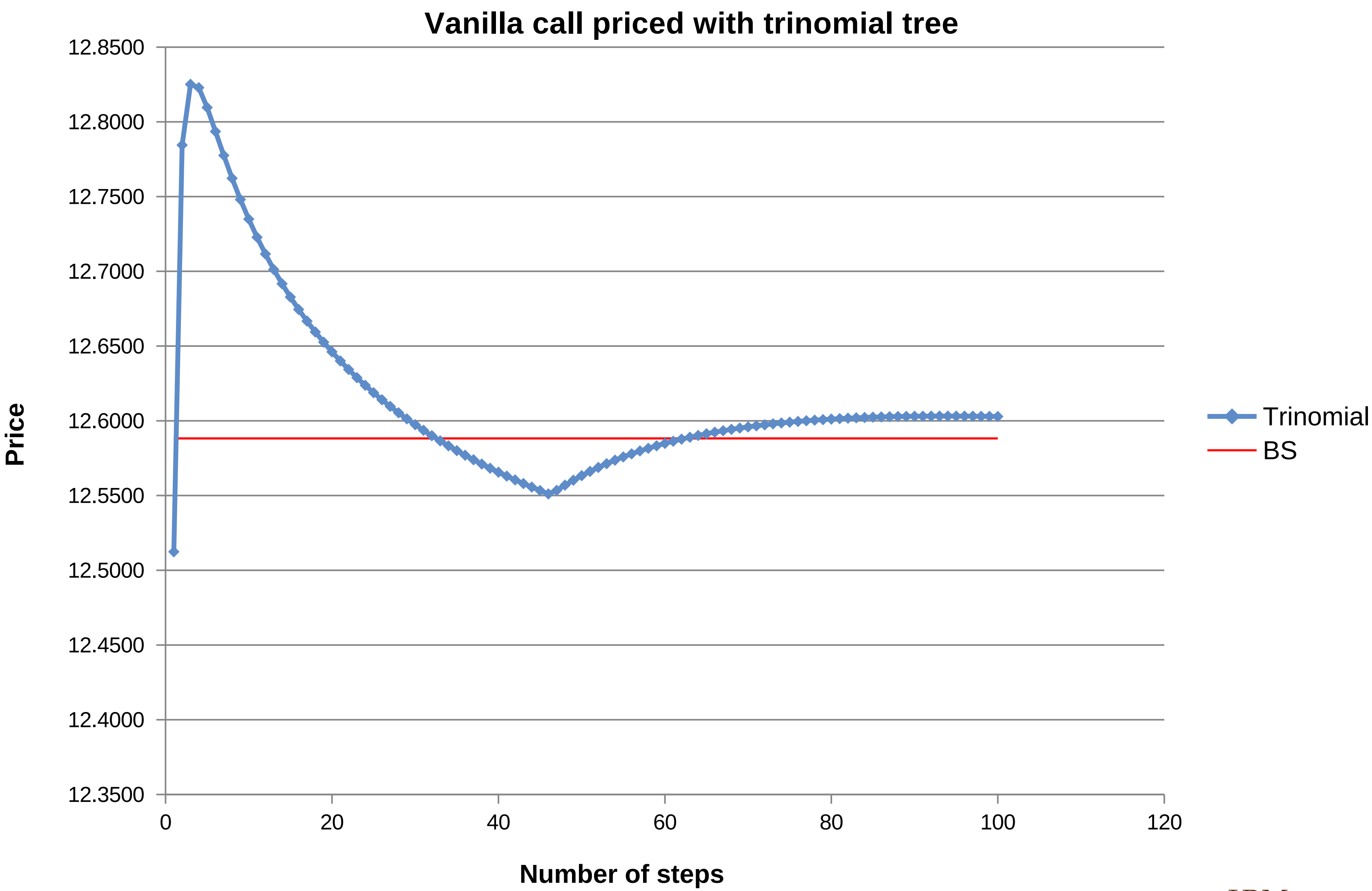
Trinomial Trees

- The simplest way to create a trinomial tree is to concatenate two time steps of a binomial tree



Trinomial Trees

- Graph of the price given by the trinomial tree vs the true price



TREES AND FINITE DIFFERENCES

Binomial vs Trinomial

- The price given by the trinomial tree does not exhibit the oscillatory pattern
- Don’t read too much into this: the trinomial tree simply picks every second price given by the binomial tree
- Because this trinomial tree with N time steps gives the same price as a binomial with $2N$ time steps

Binomial vs Trinomial

- The real advantage is speed:
  - A binomial tree with $2N$ steps has
    $1 + 2 + 3 + \cdots + 2N = 2N^2 + N$ nodes
  - But the same result can be obtained from a trinomial tree with N steps, which takes less than half the nodes:
    $1 + 3 + 5 + \cdots + (2N-1)^2 = N^2$ nodes
  - …although slightly more work has to be done for each node

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Explicit Finite Difference Schemes

- Let's take a more formal approach, starting with the Black-Scholes equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

- Switch to log-coordinates:

$$S = S_0 e^x$$

- New equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial V}{\partial x} - rV = 0$$

Explicit Finite Difference Schemes

- Discretize the variables

$$t_0 = 0, t_1 = \Delta t, t_2 = 2\Delta t, \dots, t_N = T$$

$$x_{-M} = -M\Delta x, \dots, x_{-1} = -\Delta x, x_0 = 0, x_1 = \Delta x, \dots, x_M = M\Delta x$$

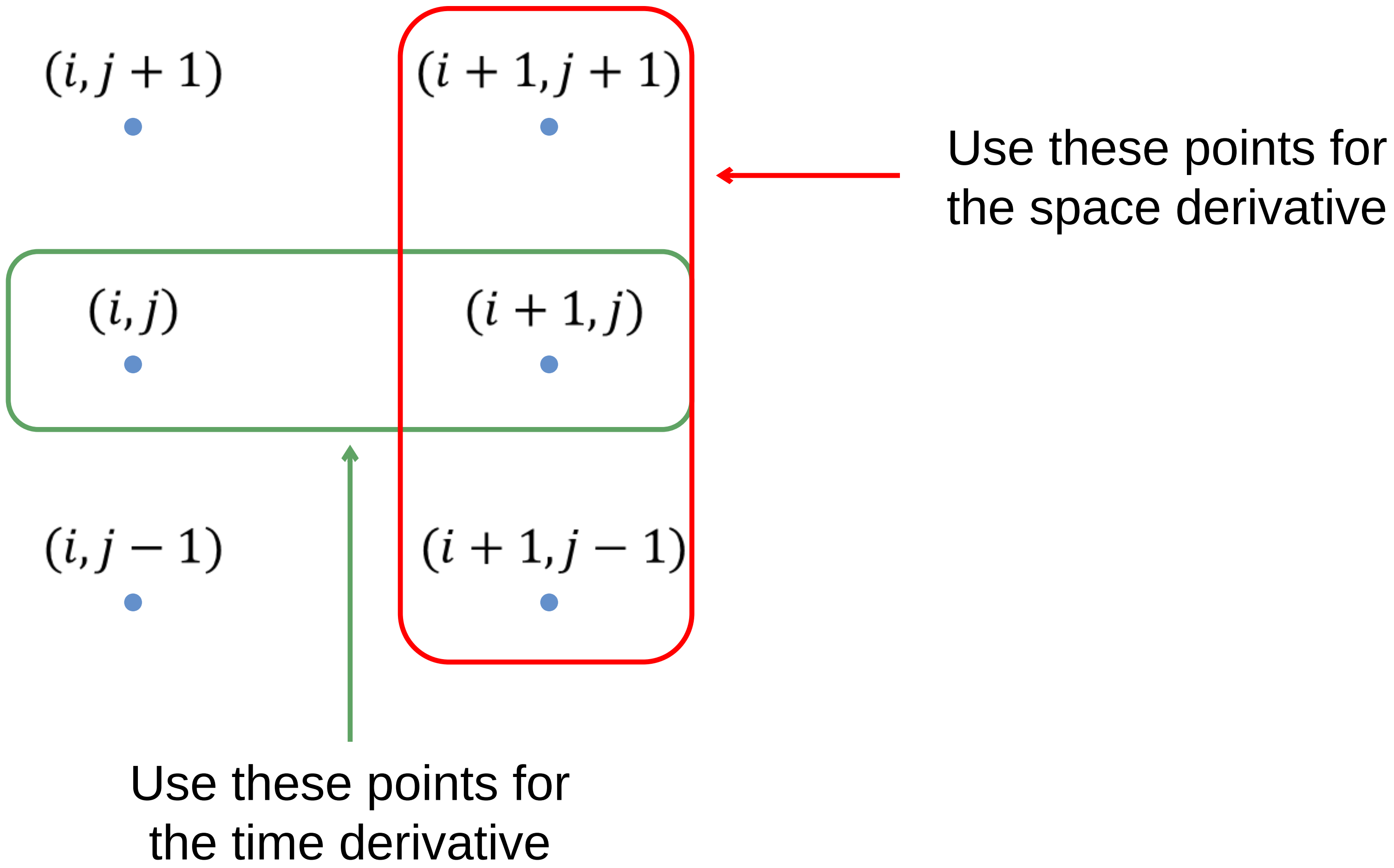
- Approximate the partial derivatives in our PDE:

$$\frac{\partial V}{\partial t}(t_i, x_j) \approx \frac{V(t_{i+1}, x_j) - V(t_i, x_j)}{\Delta t}$$

$$\frac{\partial V}{\partial x}(t_i, x_j) \approx \frac{V(t_{i+1}, x_{j+1}) - V(t_{i+1}, x_{j-1})}{2\Delta x}$$

$$\frac{\partial^2 V}{\partial x^2}(t_i, x_j) \approx \frac{V(t_{i+1}, x_{j+1}) - 2V(t_{i+1}, x_j) + V(t_{i+1}, x_{j-1})}{(\Delta x)^2}$$

Explicit Finite Difference Schemes



Explicit Finite Difference Schemes

- This allows us to calculate $V(t_i, x_j)$ using only the values of V at the next time-step t_{i+1}:

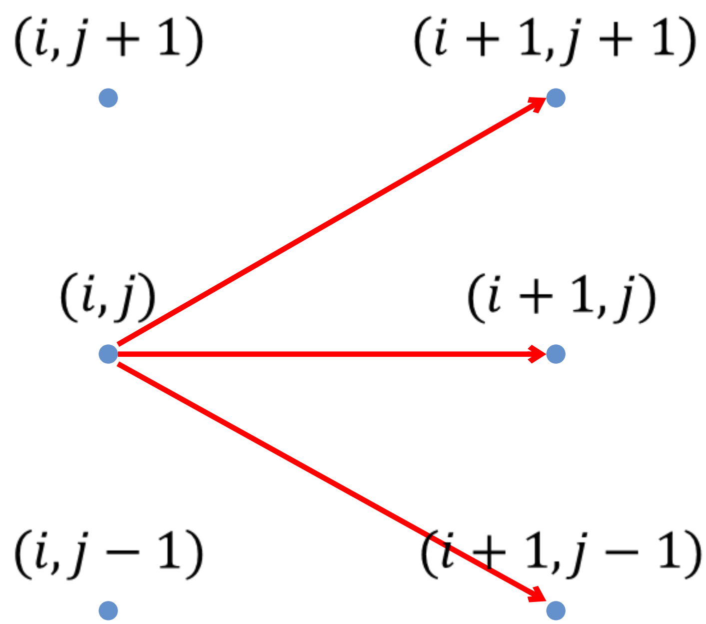
$$V(t_i, x_j) = \left(\frac{\sigma^2 \Delta t}{2(\Delta x)^2} + \left(r - \frac{\sigma^2}{2}\right)\frac{\Delta t}{2\Delta x}\right) \cdot V(t_{i+1}, x_{j+1})$$

$$+ \left(1 - \frac{\sigma^2 \Delta t}{(\Delta x)^2} + r\Delta t\right) \cdot V(t_{i+1}, x_j)$$

$$+ \left(\frac{\sigma^2 \Delta t}{2(\Delta x)^2} - \left(r - \frac{\sigma^2}{2}\right)\frac{\Delta \mathrm{t}}{2\Delta x}\right) \cdot V(t_{i+1}, x_{j-1})$$

Explicit Finite Difference Schemes

- In other words this scheme produces a trinomial tree
- The node (i,j) is calculated in terms of the three nodes branching from it:



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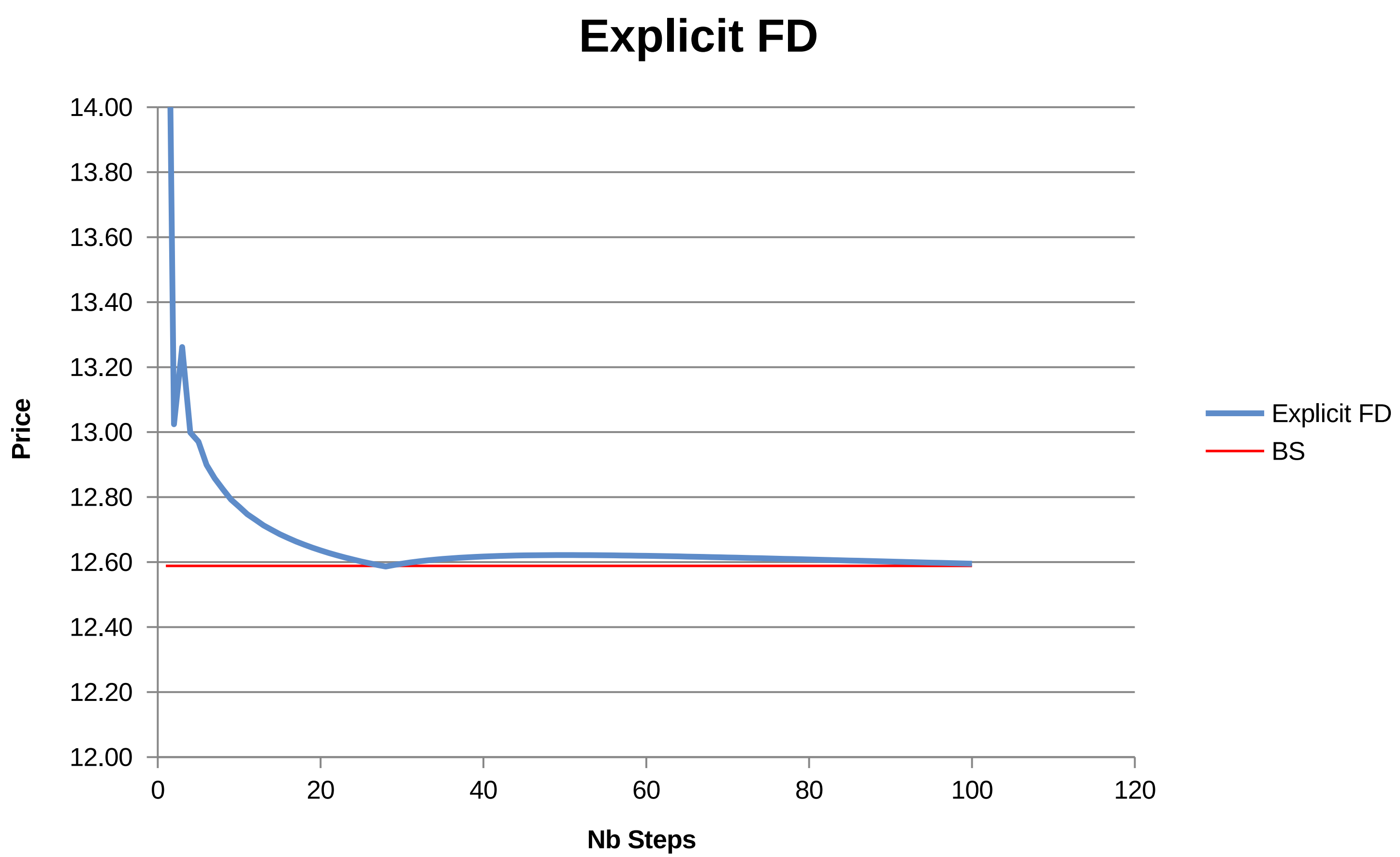
Explicit Finite Difference Schemes

- Graph of the price given by the explicit scheme vs the true price (same parameters as for the trees)

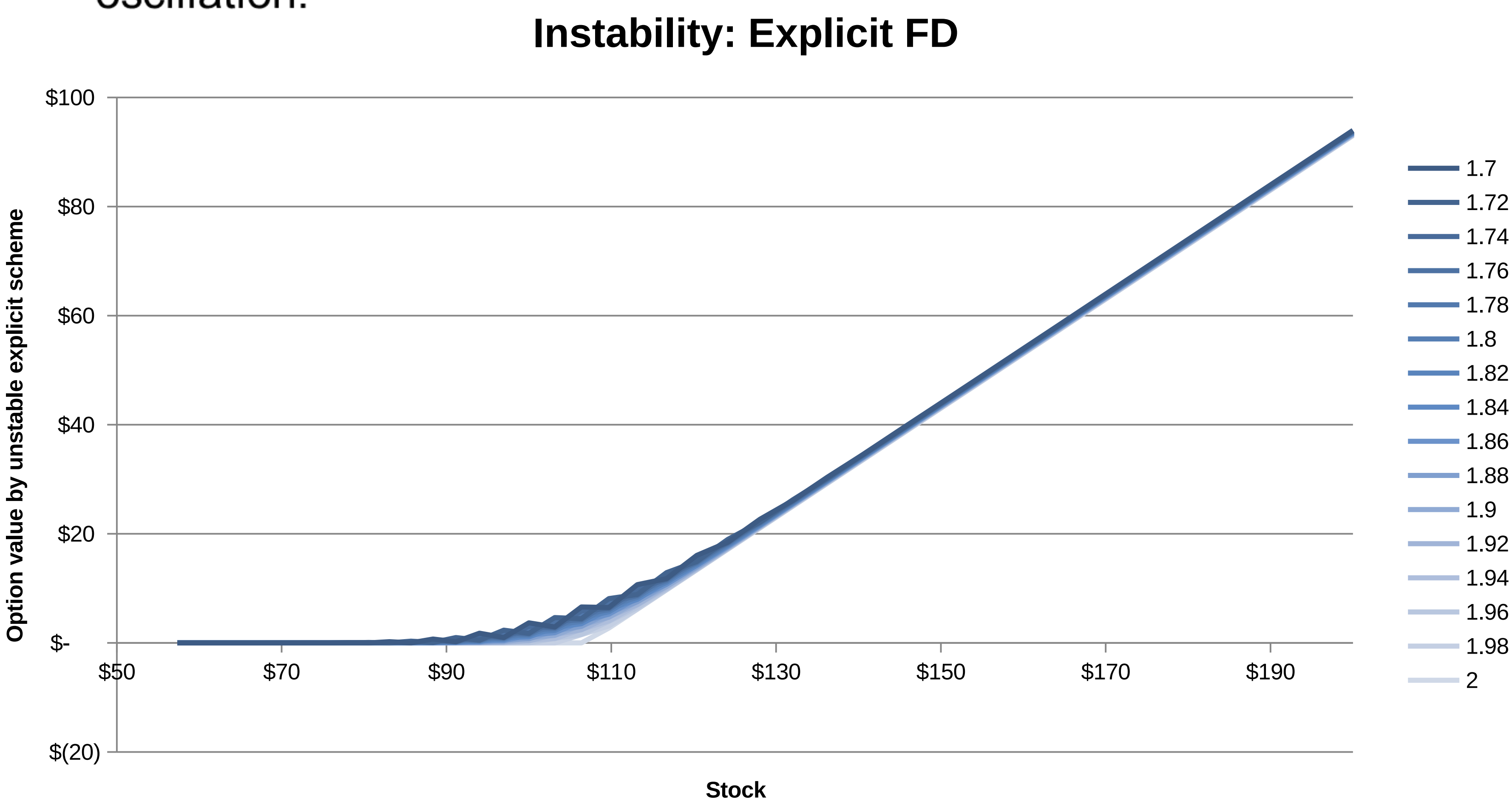


Explicit Finite Difference Schemes

- The explicit scheme is only conditionally stable
- Instability means that errors get amplified
  - This happens when the time step is too large compared to the space discretization
  - Roughly, we need $\sigma^2 \Delta t \leq (\Delta x)^2$ for stability
- For example take a look at the s/s “Finite differences”, tab “Explicit – Instability”
- We show the graph of the calculated value of the option at successive time steps

Explicit Finite Difference Schemes

- As the time gets farther from maturity ($t = 2$) we start to see some oscillation:



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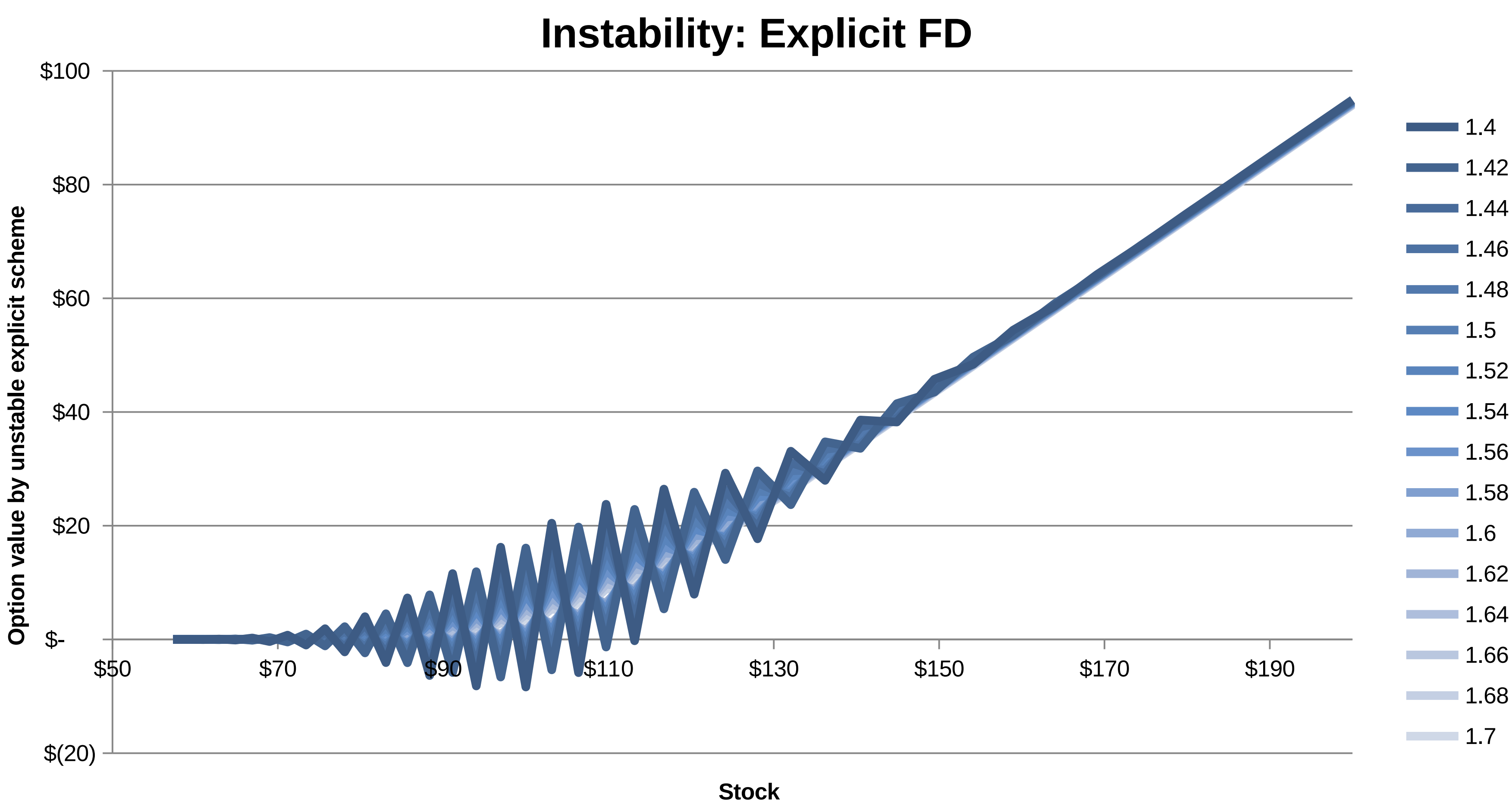
Explicit Finite Difference Schemes

- Further away from maturity the instability becomes huge



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- *Implicit & Crank-Nicolson*
- Barrier Options
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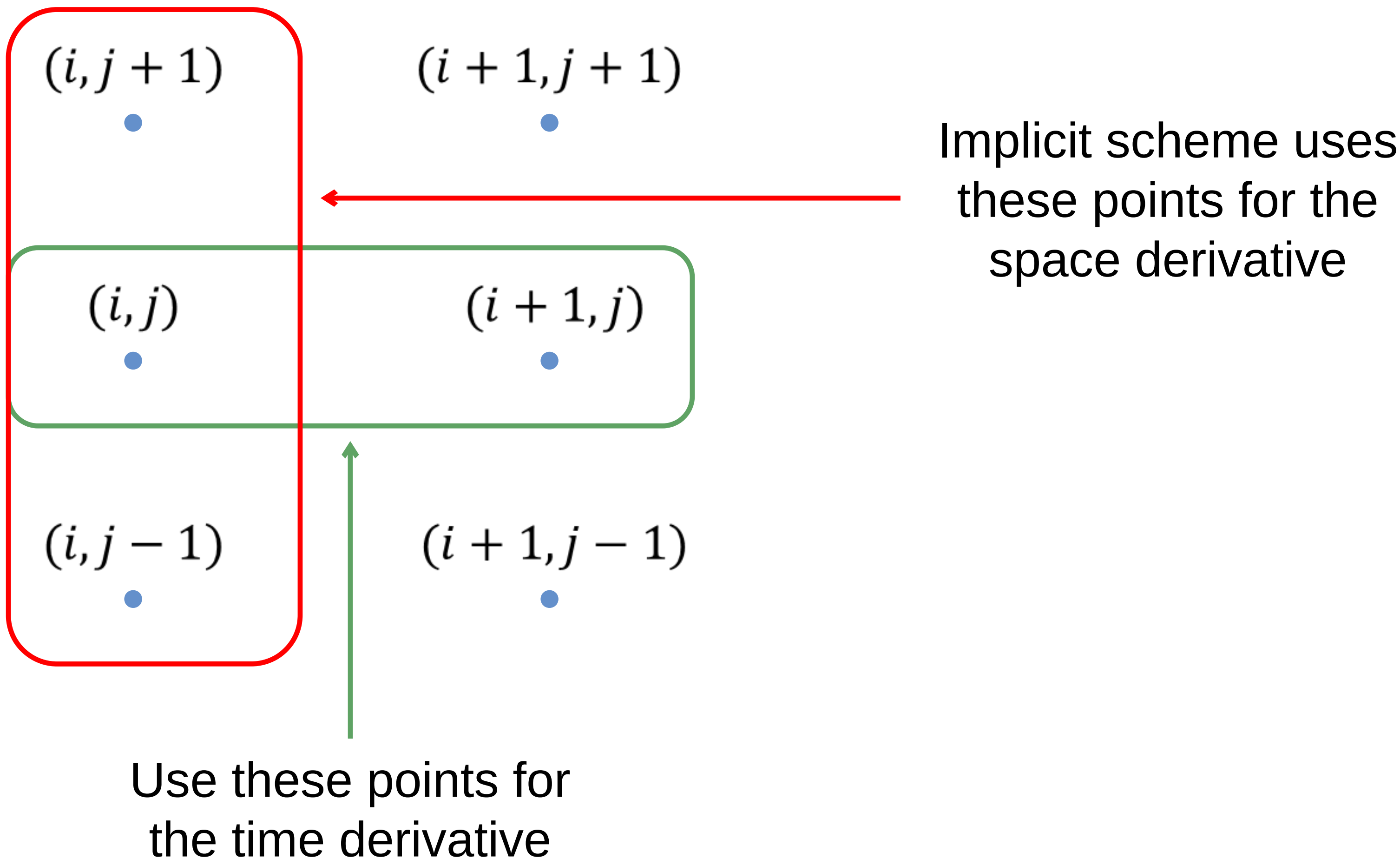
Implicit & Crank-Nicolson

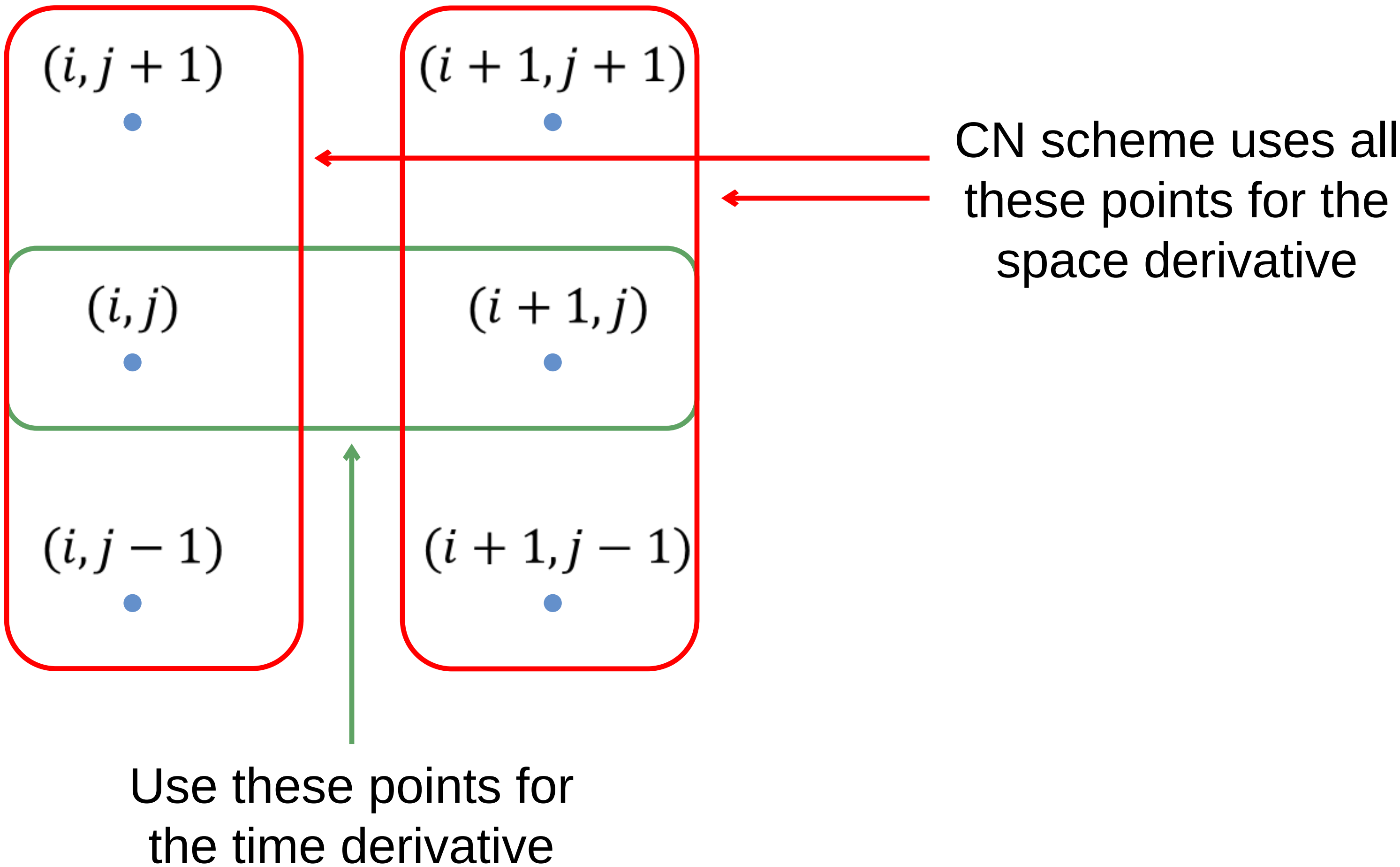
- The implicit and CN schemes were designed to fix the instability problem of the explicit scheme
- They are slightly more complicated to implement: they require to solve a linear system when we do the backward induction
- But are unconditionally stable*
- And CN converges faster: the first-order time discretization error cancels out, making it second-order in both time and space

*In CN, high-frequency oscillations are not damped: non-smooth payoffs (i.e., almost all real products) can lead to “ringing” effects

Implicit & Crank-Nicolson

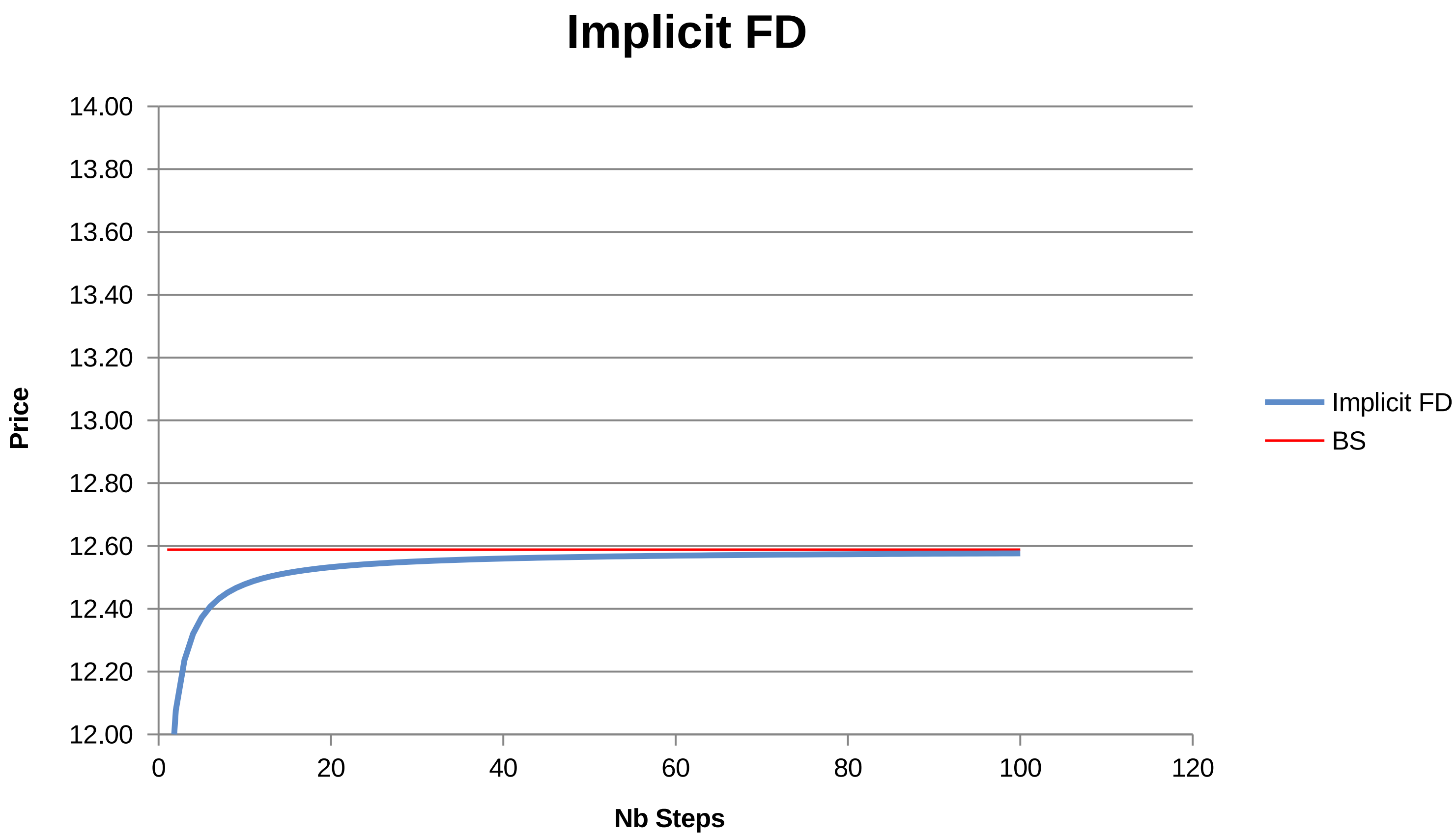


Implicit & Crank-Nicolson



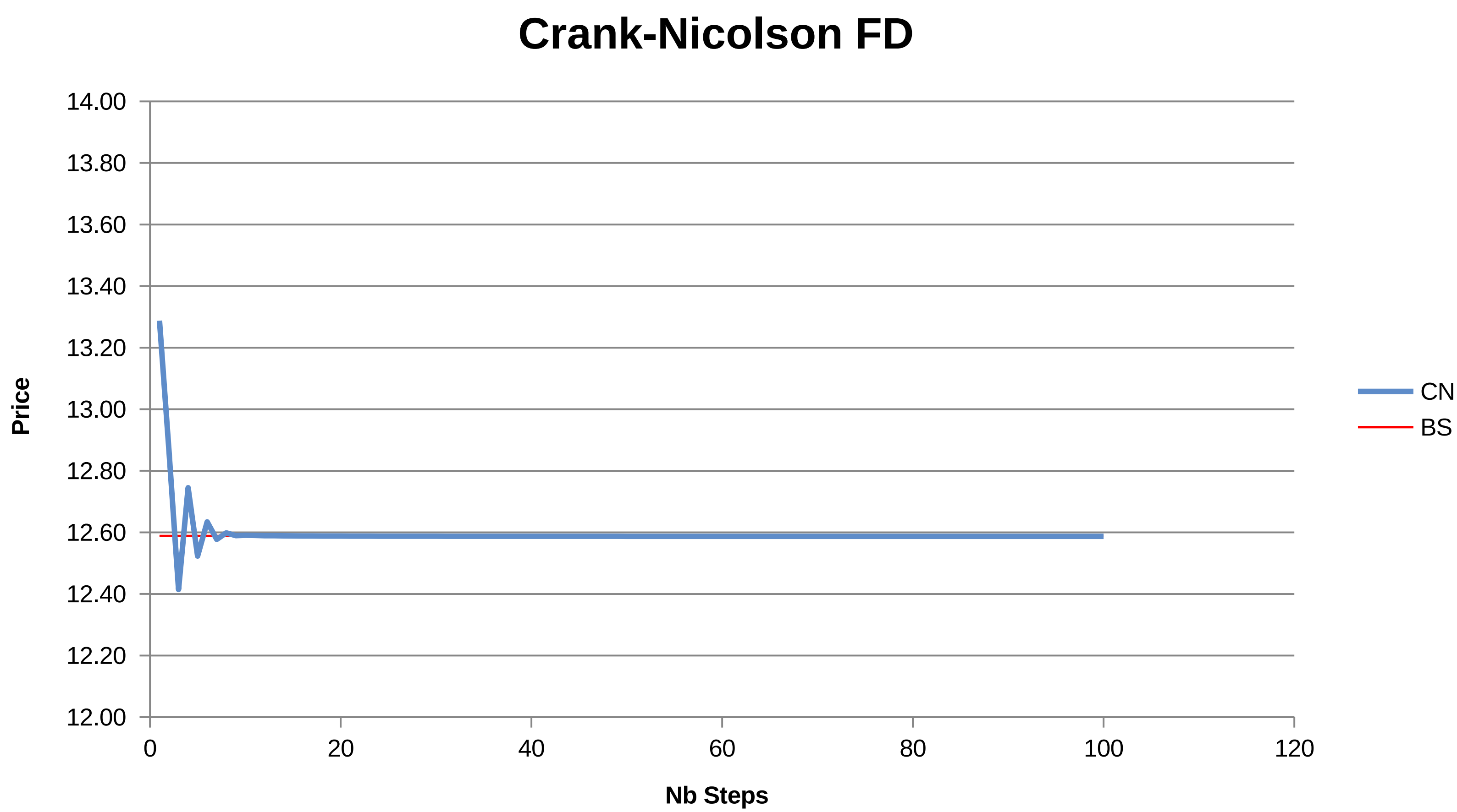
Implicit & Crank-Nicolson

- Graph of the option price calculated with an implicit scheme vs the true price



Implicit & Crank-Nicolson

- Graph of the option price calculated with Crank-Nicolson vs the true price



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Barrier Options

- Reflection principle for Brownian motion
- Let τ_m be the first time Brownian reaches level m
- We can show that
- $\mathbb{P}\{\tau_m \leq t, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$
- Set w = m in above equation and we have,
- $\mathbb{P}\{\tau_m \leq t, W(t) \leq m\} = \mathbb{P}\{W(t) \geq m\}$
- Now let's try to compute $\mathbb{P}\{\tau_m \leq t\}$
- $\mathbb{P}\{\tau_m \leq t\} = \mathbb{P}\{\tau_m \leq t | w(t) > m\} + \mathbb{P}\{\tau_m \leq t | w(t) < m\}$
- $\mathbb{P}\{\tau_m \leq t\} = 2\mathbb{P}\{W(t) \geq m\}$
- We note that $\tau_m \leq t$ is equivalent to $M(t) \geq m \; where \; M(t) = \max_{0 \leq s \leq t} w(s)$

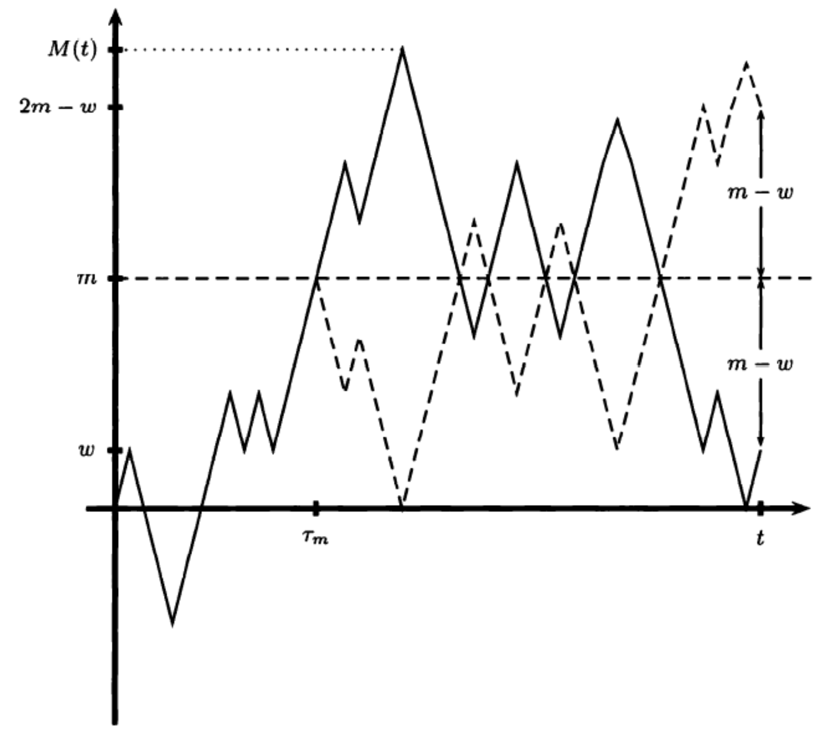

Fig. 3.7.1. Brownian path and reflected path.

Barrier Options

- $\mathbb{P}\{M(t) \geq m, W(t) \leq w\} = \mathbb{P}\{W(t) \geq 2m - w\}$
- Now we can derive joint distribution of M and W.

$$\mathbb{P}\{M(t) \geq m, W(t) \leq w\} = \int_m^\infty \int_{-\infty}^w f_{M(t),W(t)}(x,y)\,dy\,dx$$

and

$$\mathbb{P}\{W(t) \geq 2m - w\} = \frac{1}{\sqrt{2\pi t}} \int_{2m-w}^\infty e^{-\frac{z^2}{2t}}\,dz,$$

we have from (3.7.6) that

$$\int_m^\infty \int_{-\infty}^w f_{M(t),W(t)}(x,y)\,dy\,dx = \frac{1}{\sqrt{2\pi t}} \int_{2m-w}^\infty e^{-\frac{z^2}{2t}}\,dz.$$

We differentiate first with respect to m to obtain

$$-\int_{-\infty}^w f_{M(t),W(t)}(m,y)\,dy = -\frac{2}{\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}.$$

We next differentiate with respect to w to see that

$$-f_{M(t),W(t)}(m,w) = -\frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-\frac{(2m-w)^2}{2t}}.$$

Barrier Options

- Path-dependent options are difficult to implement in a finite difference scheme
- But barrier options are only mildly path-dependent
- It’s very simple to modify a scheme to price a knock-out barrier option: simply set the option price to zero for all the nodes beyond the barrier
- To price a knock-in you can either:
  - Use KI-KO parity; or
  - Price a vanilla on an identical grid, and use the vanilla option value as the boundary condition for the KI option

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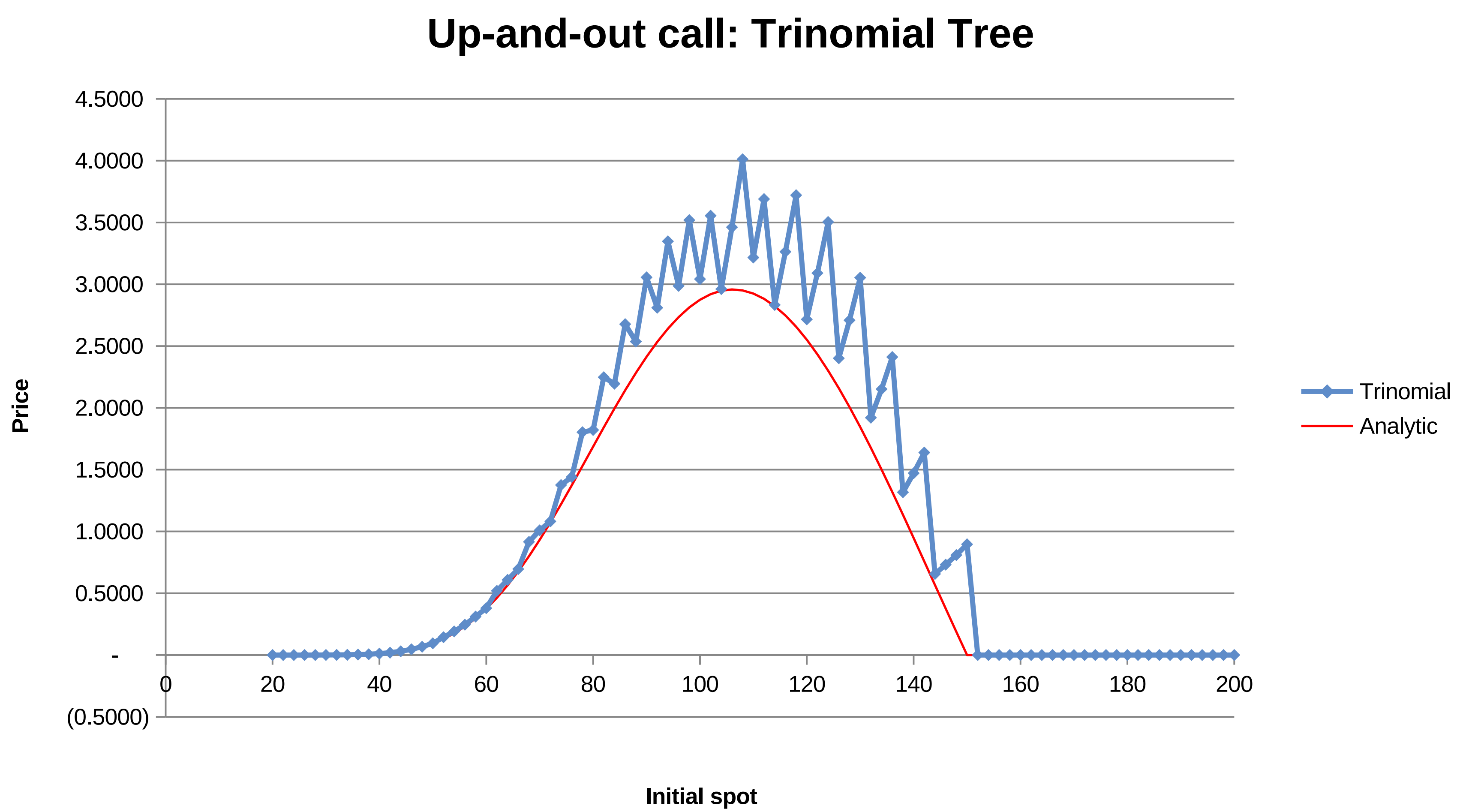
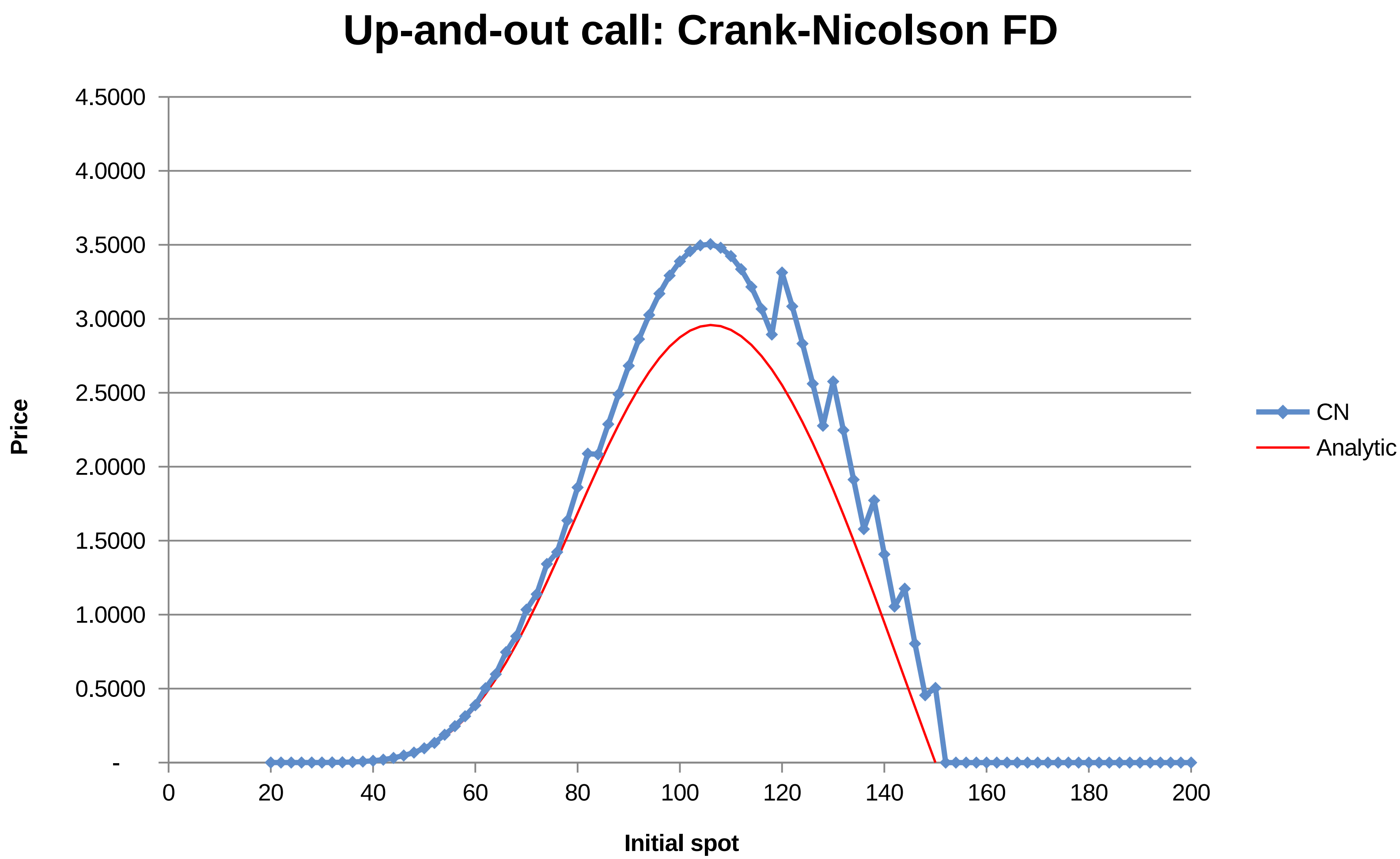
Barrier Options

- However, this straightforward implementation runs into trouble:



Barrier Options

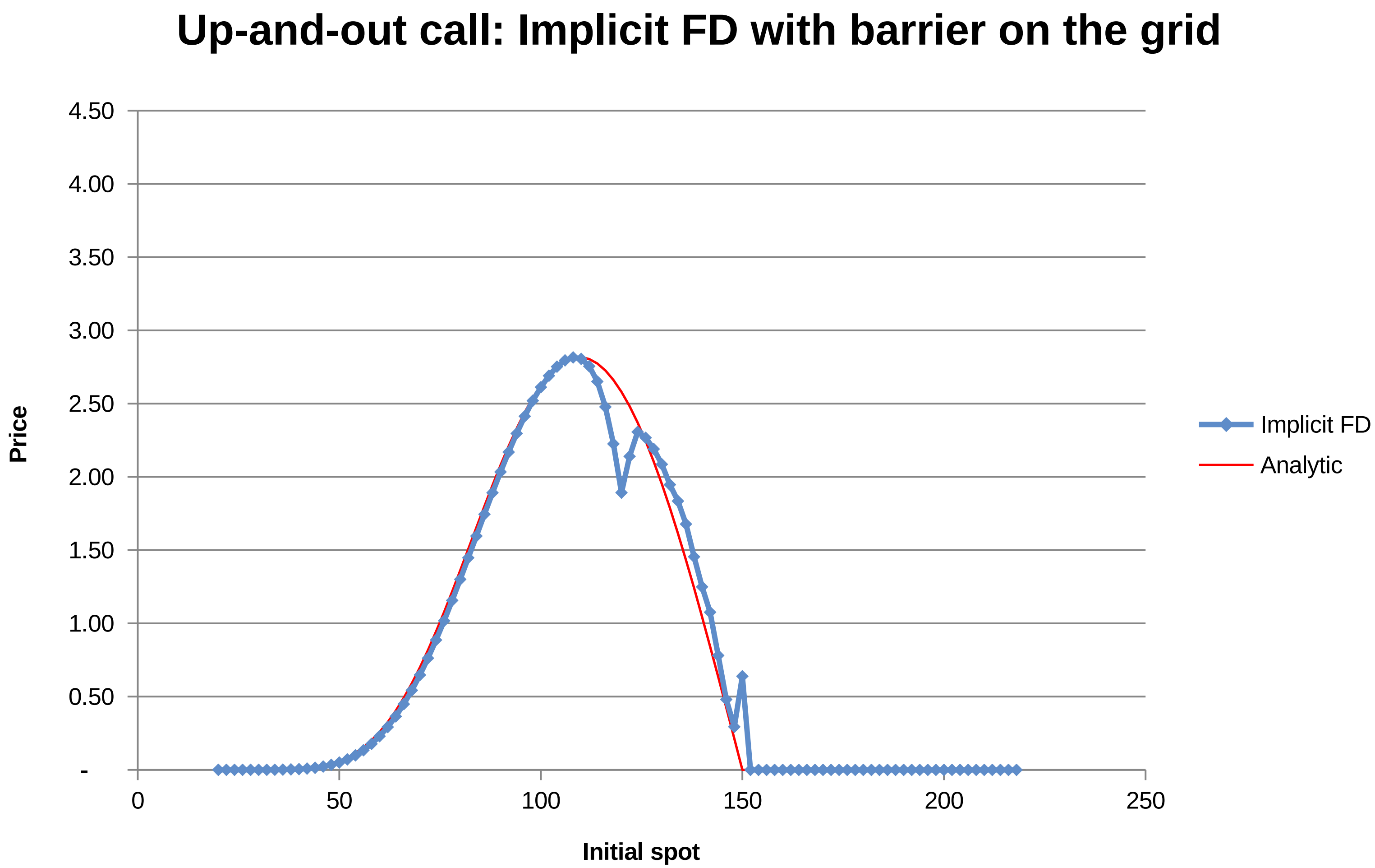
- Crank-Nicolson is not much better:



Barrier Options

- The reason is that the barrier option is discontinuous
- One way to fix this is to make sure the barrier is among the grid points of your tree or FD scheme
- In the next slide we show an implicit scheme where the barrier level is on the grid
- The improvement is clear (there is still some error due to too few points being used)

Barrier Options



Barrier Options

- The improvement is clear, but there is still some noise
- There are other tricks we can use…

Trees and FD vs Monte Carlo

- Trees and FD advantages:
  - Fast convergence
  - Very easy to implement American optionality
- Trees and FD disadvantages:
  - Slow in higher dimensions, impractical for dim > 4
  - Difficult to use for path-dependent options

Trees and FD vs Monte Carlo

- Monte Carlo advantages:
  - Very easy to implement
  - Works well in higher dimensions
  - Perfect for path-dependent options
- Monte Carlo disadvantages:
  - Slow convergence
  - Difficult to implement American optionality